Karman Sidhu

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EDUCATION

McMaster University – Bachelor of Engineering (Co-op) *Hamilton, Ontario*
Mechatronics Engineering *Expected Graduation: April 2029*

Technical Coursework: Embedded Systems Design I, Data Structures & Algorithms, Analog & Digital Circuits, Mechanics (Dynamics & Statics), Mechanical Engineering Measurements

EXPERIENCE

Data Clerk (Contract, Full-Time) – TJX Canada *May 2025 – Aug. 2025*

- Processed and validated 300+ SCAN operational data daily under strict deadlines, ensuring data accuracy and quality control
- Analyzed data inconsistencies and errors to support system integrity, troubleshooting, and process reliability
- Collaborated with supervisors and cross-functional teams to investigate recurring issues and improve data consistency and workflow efficiency.
- Utilized Microsoft Excel and internal database system for data verification, reporting, and record management in a fast-paced environment.

Sales Associate (Part-Time) Adidas *Oct. 2023 – Jan. 2026*

- Delivered customer service in a high-volume, KPI-driven retail environment, while supporting customer satisfaction and meeting operational targets
- Operated POS system and supported store operations including transactions, product inquiries, onboarding assistance, and team workflow during peak periods.

SKILLS

Programming: Java, Python, C++, C, HTML, CSS, JavaScript, Embedded Systems
Software: SolidWorks, Autodesk Inventor, MATLAB, Arduino, Git, Microsoft Office (Excel, PowerPoint)
Manufacturing: Machining (mill, lathe), carbon fiber composites, soldering, hand/power tools

LEADERSHIP & INVOLVEMENT

McMaster Engineers Without Borders – Internal Team/Conference Co-Lead *Sept. 2024 – Present*

- Co-organize MAC-IMPACT conference logistics, planning, operations, and speaker outreach to multiple universities
- Led a technical workshop on water treatment systems with the EWB Toronto Professional Chapter
- Coordinate and facilitate weekly informational sessions, technical workshops, and charity drives

MAC Formula Electric – Manufacturing Team *Sept. 2024 – Sept. 2025*

- Supported chassis fabrication (carbon fiber layups, resin infusion, structural assembly)
- Machined parts (e.g. mounts), supported aerodynamic and accumulator cart manufacturing

PROJECTS

Adaptive PID Flow Control System *May 2026 – June 2026*

- PID control system using Arduino Uno (C++), servo motor (PWM), and ultrasonic sensor feedback to simulate obstruction-aware flow management via Mealy FSM, SPI protocols & CAD modeling

Maintry – Personal Asset Maintenance Platform *Feb. 2026 – May 2026*

- Developed a full-stack web application using Node.js, MongoDB, and JavaScript to track maintenance history, manage assets, schedule services, analyze costs/usage patterns
- Integrates Google Places API to fetch, filter, and save professionals with ratings and contact details.

Sequential Digital Circuit *Nov. 2025 – Dec. 2025*

- Designed a sequential logic FSM using JK flip-flops and Karnaugh maps to cycle a digit sequence on a 7-segment display via BCD decoding and Multisim simulation.

Additional Projects: Adaptive Sleep Audio (PWM, TMRpcm) | **Thermal Regulator** (LM35, MOSFET, FSM)